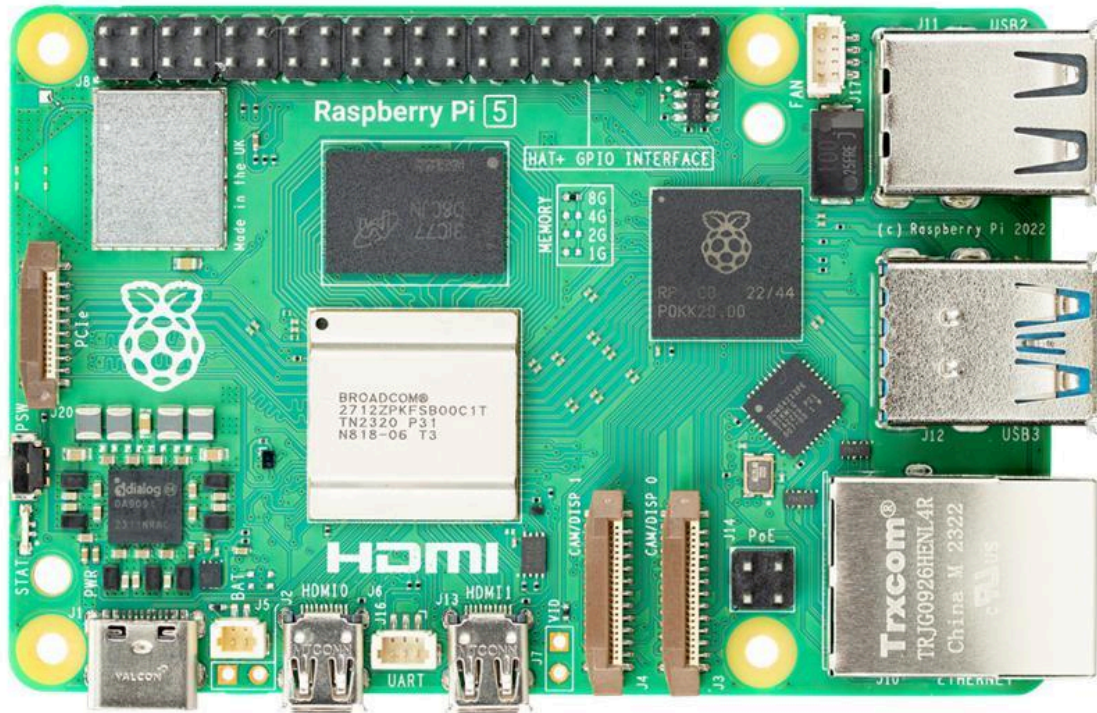
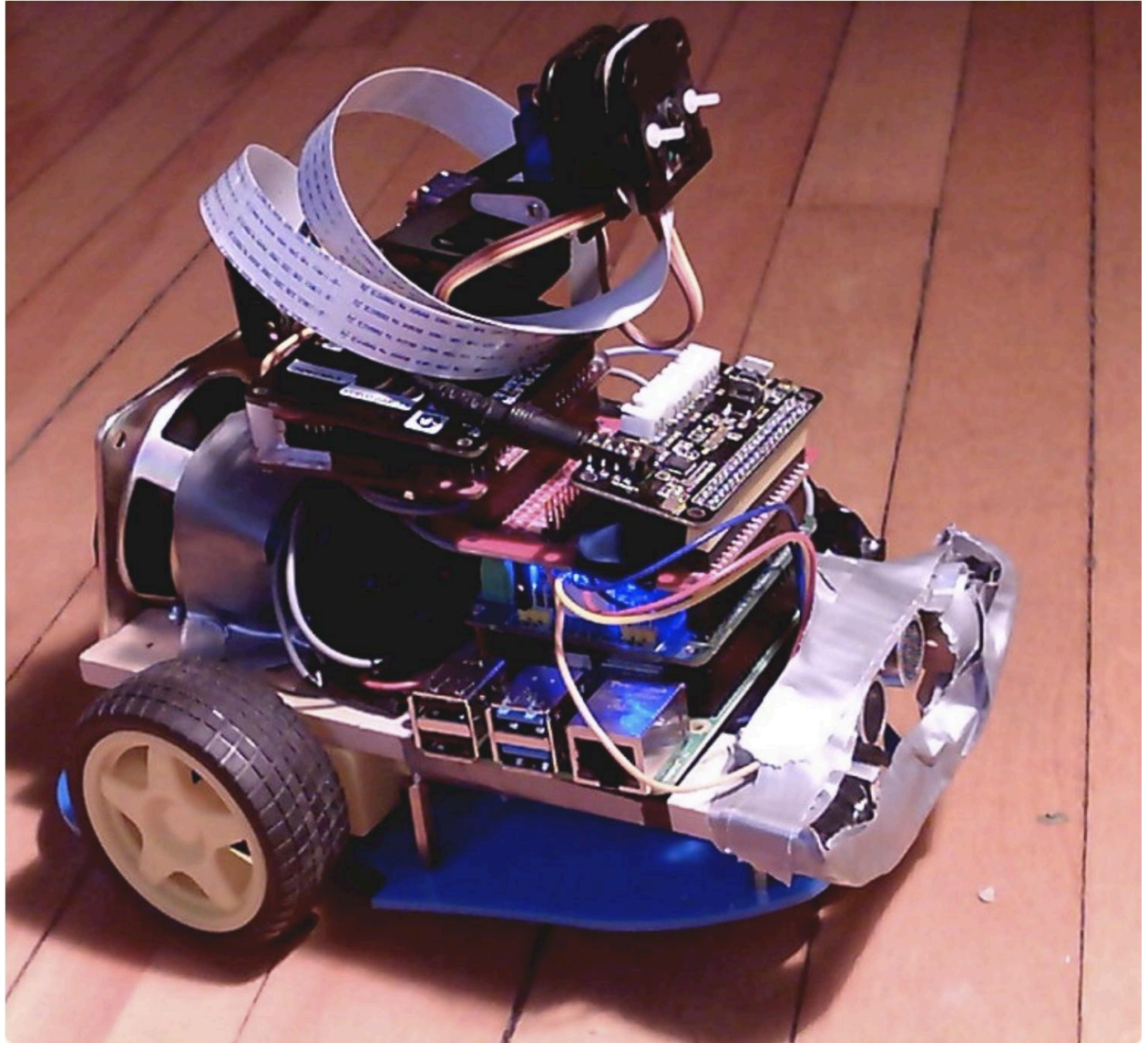
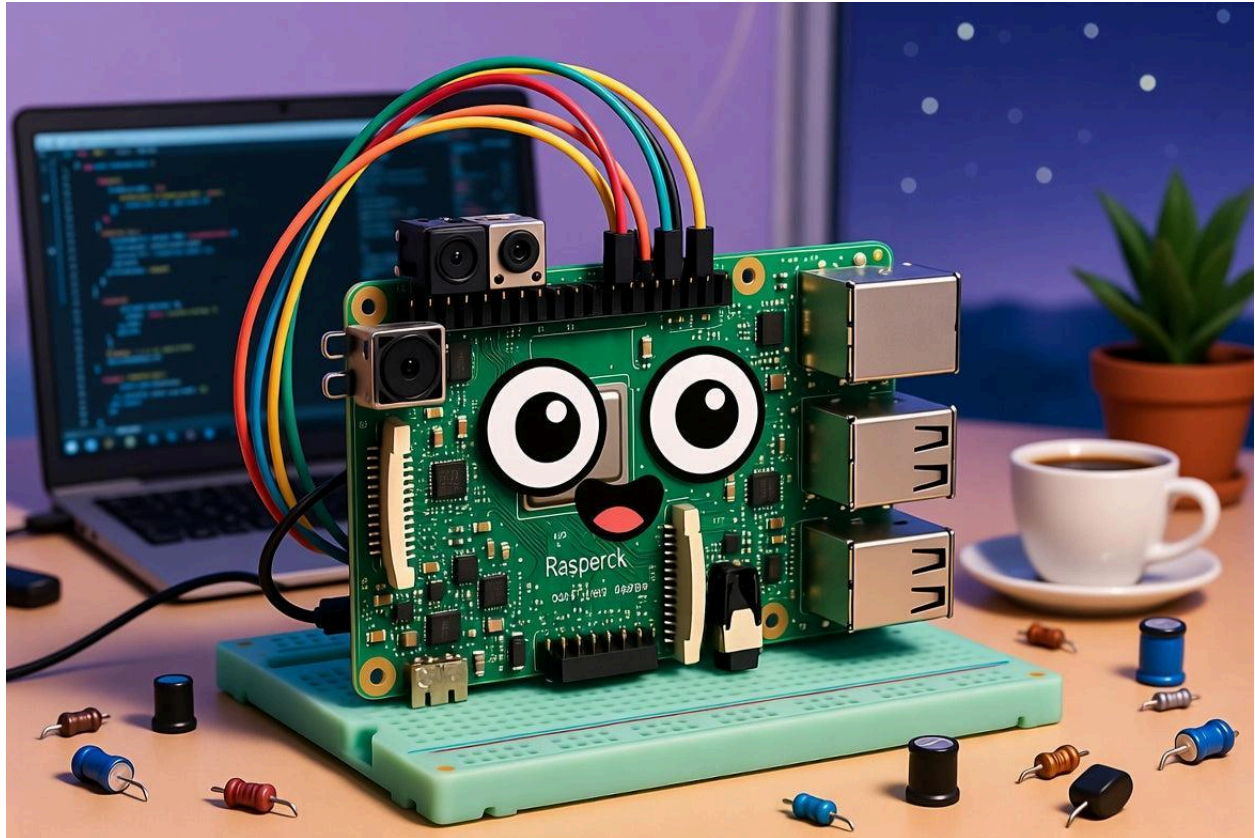


Powerful Embedded Hardware Stack

1. Raspberry Pi









Setting up a Machine Learning Environment on Raspberry Pi

Choosing the Raspberry Pi model



Uses

- AI projects
- Robotics
- CCTV systems
- Servers

- Drone control

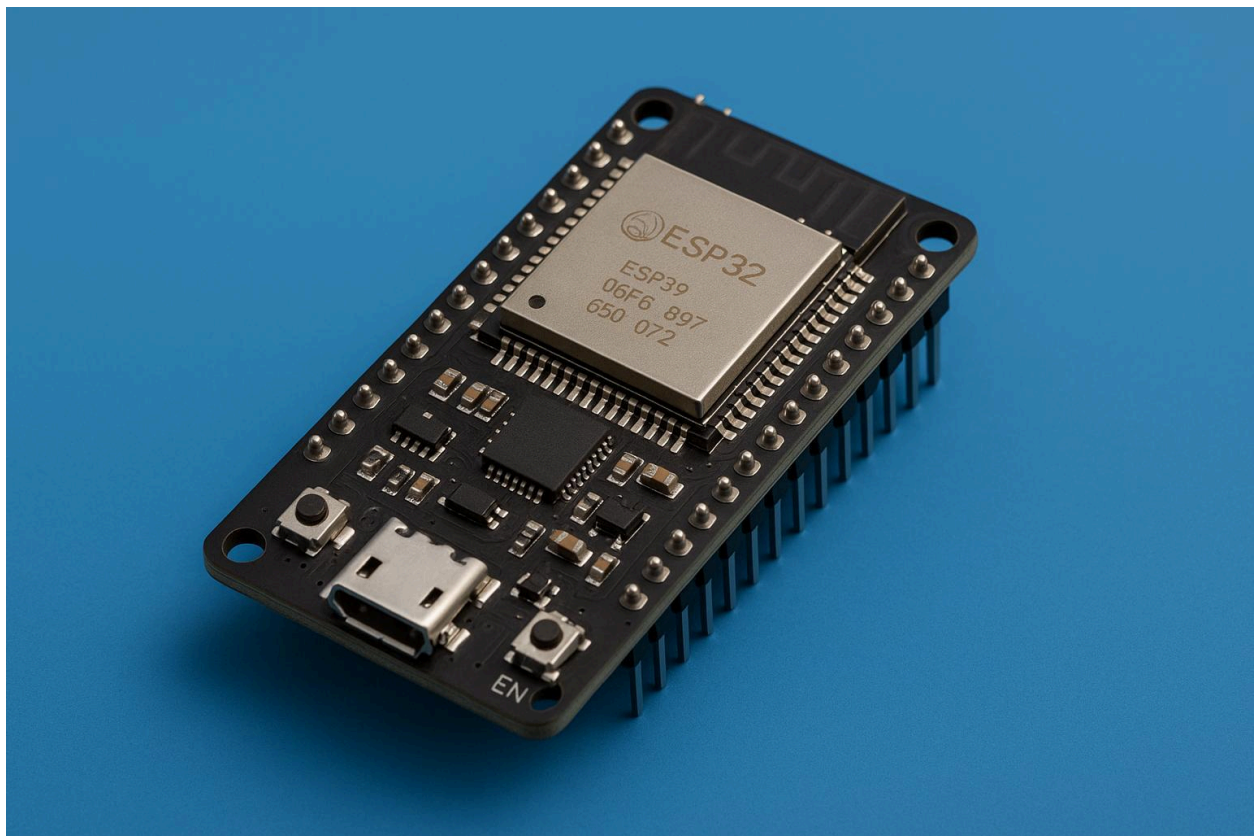
Best For

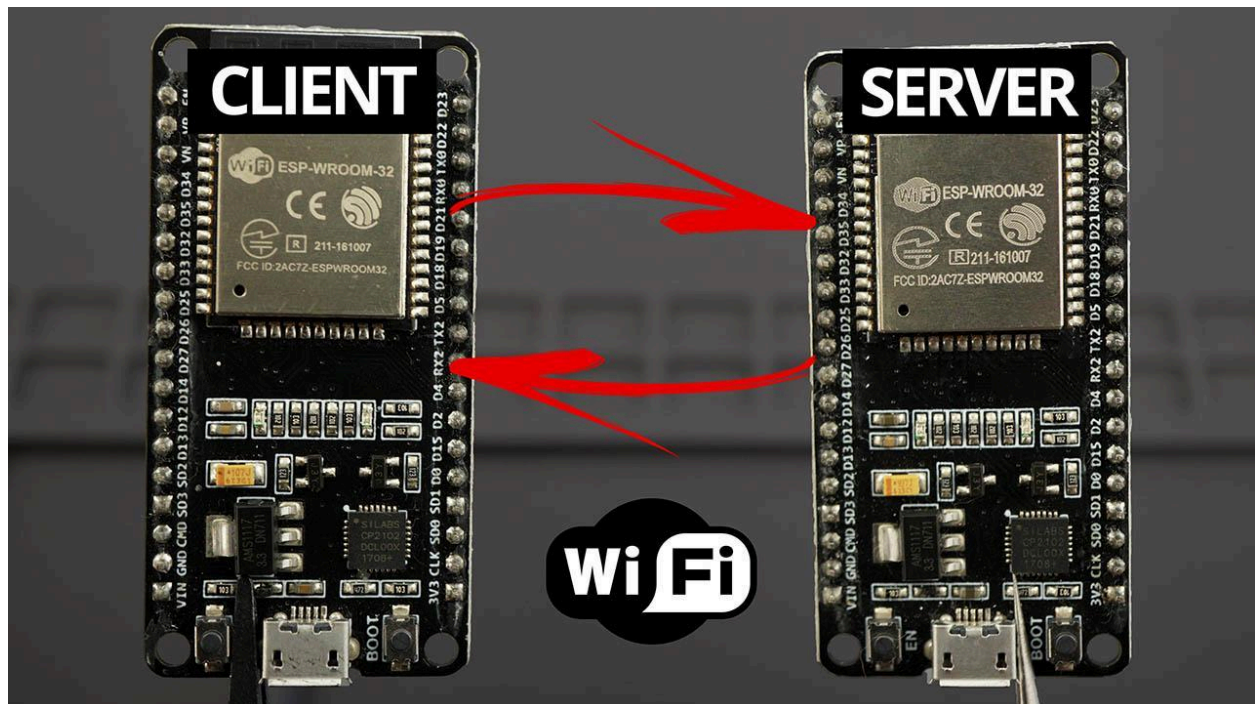
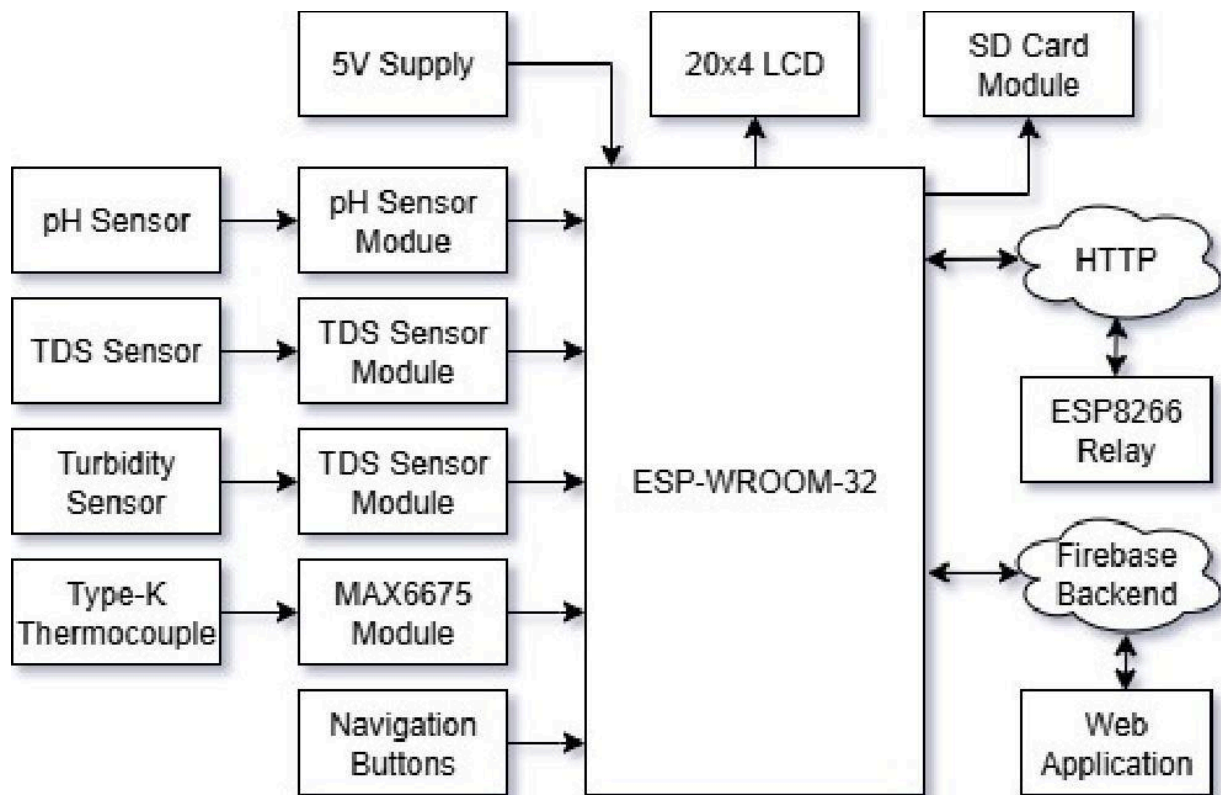
- Python
- OpenCV
- Linux
- Networking

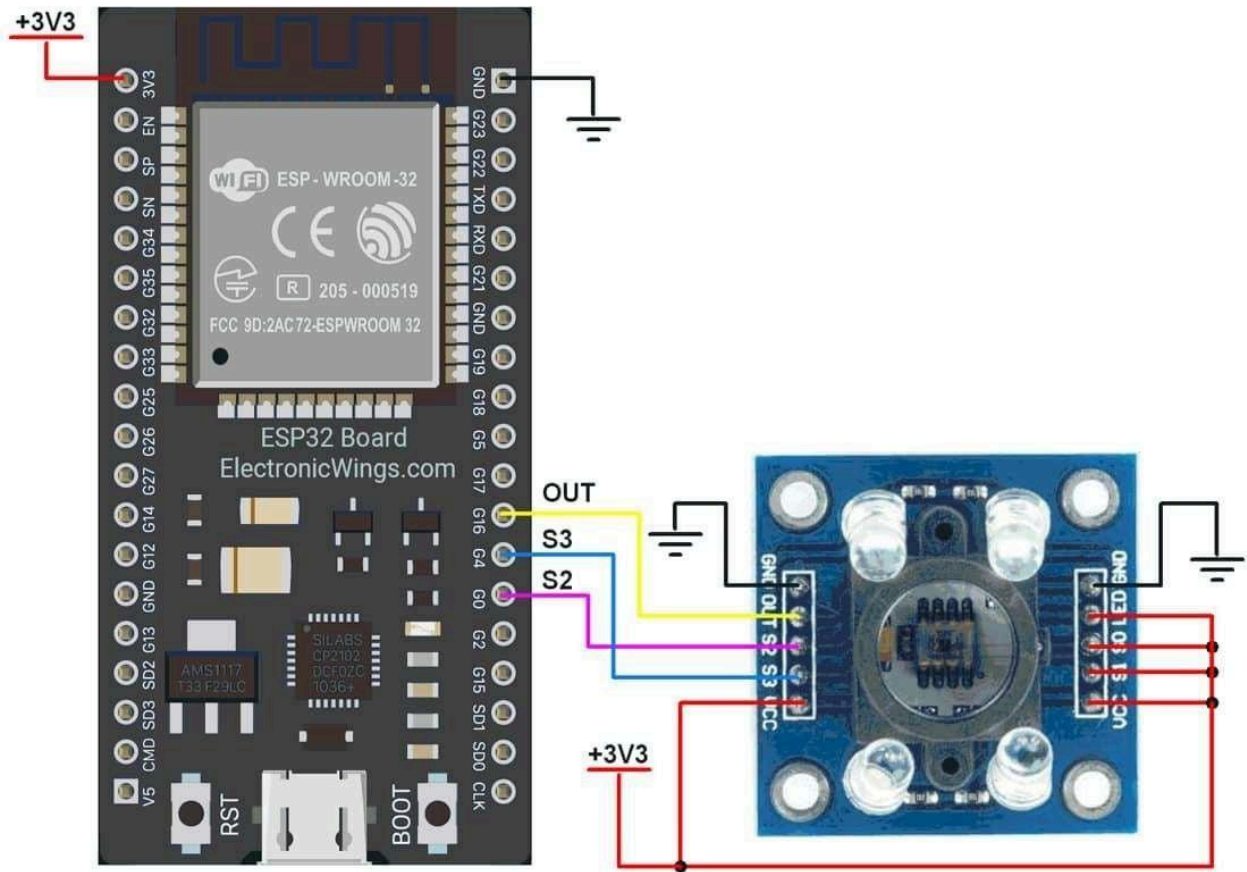
Popular Models

- Raspberry Pi 4
 - Raspberry Pi 5
 - Raspberry Pi Zero
-

2. ESP32









Features

- Built-in Wi-Fi
- Bluetooth
- Low power
- Dual core MCU

Uses

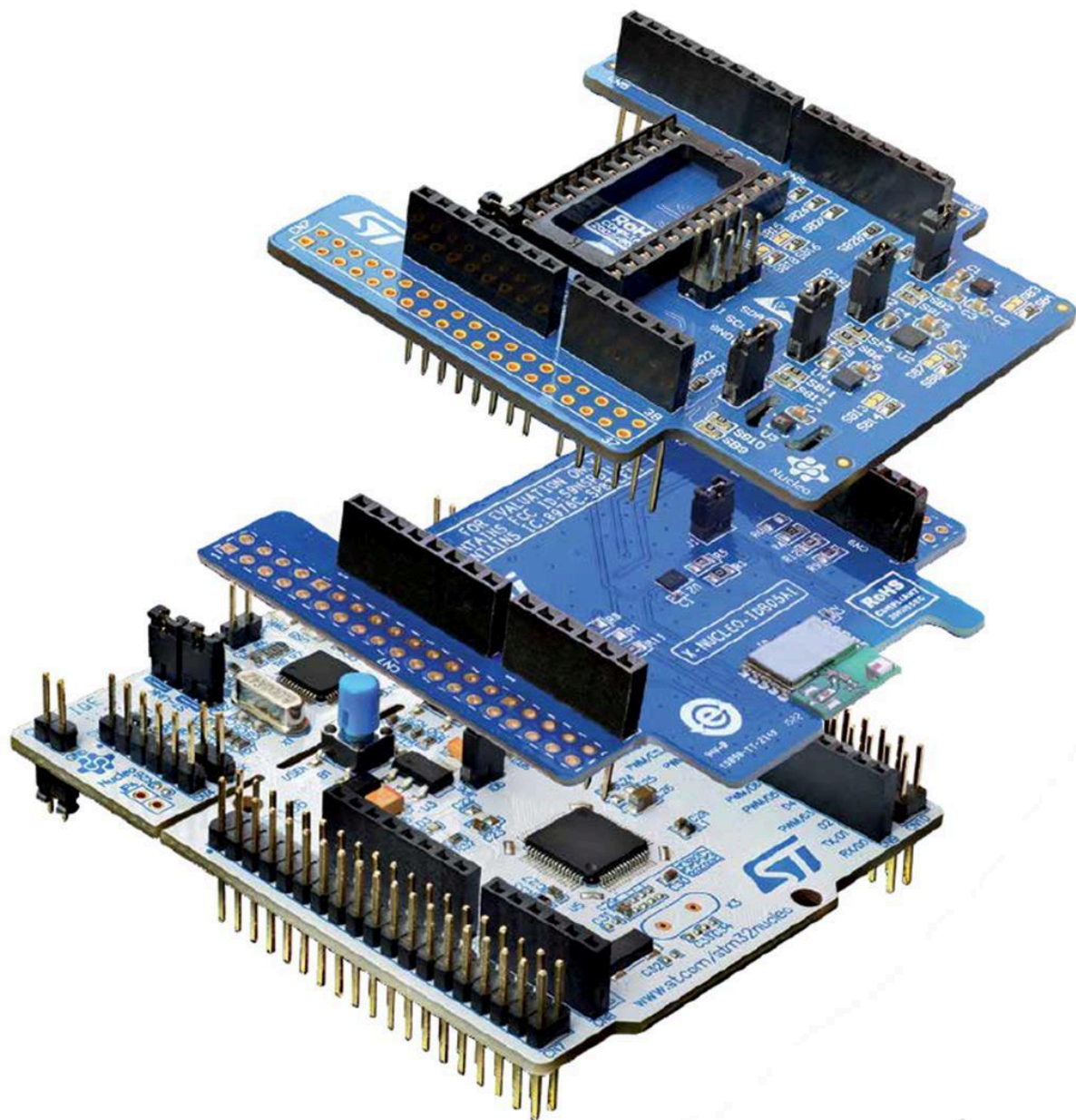
- IoT

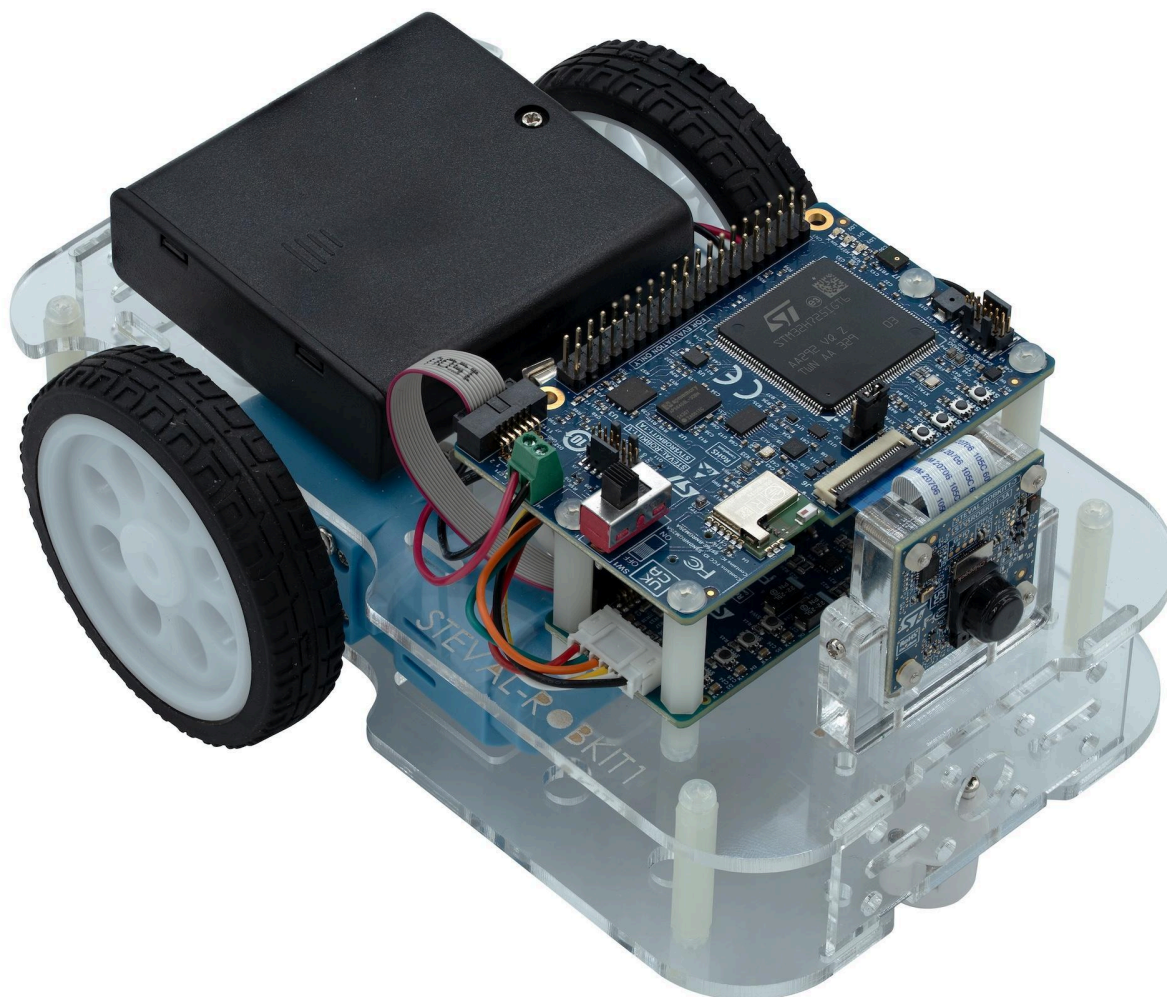
- Sensor networks
- Smart devices
- Wireless communication

Programming

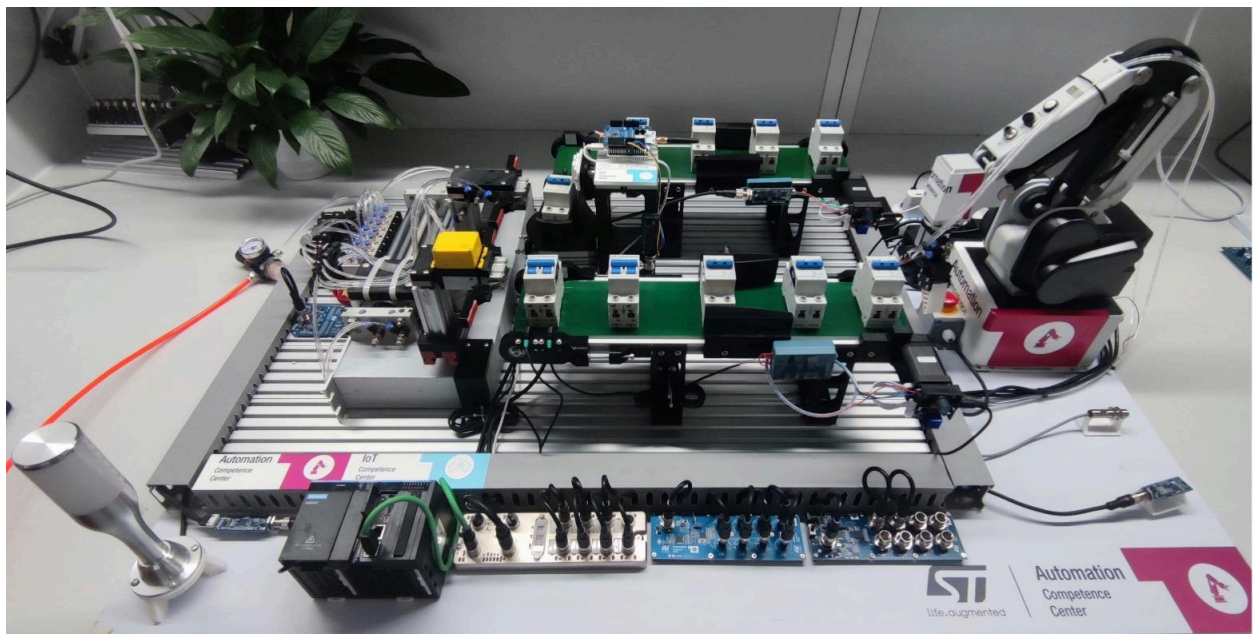
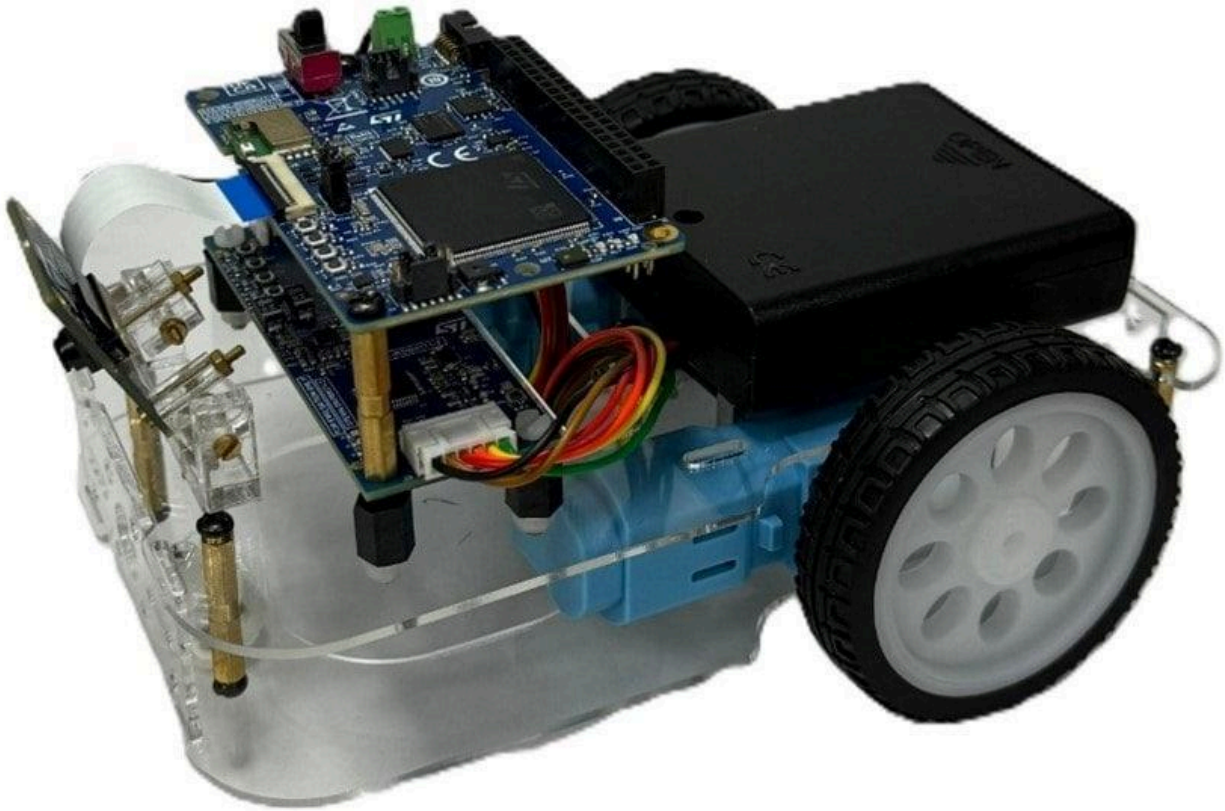
- Arduino IDE
 - MicroPython
 - ESP-IDF
-

3. STM32









Best For

- Real-time systems

- Industrial automation
- Robotics
- Defense systems

Advantages

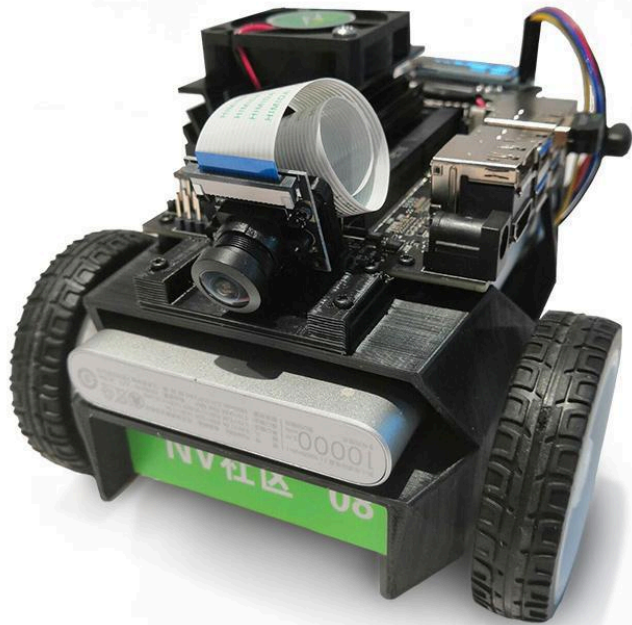
- Very fast
- Low latency
- Professional-grade MCU

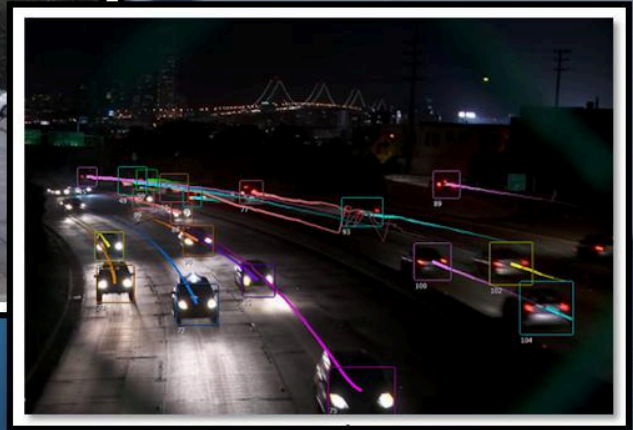
Languages

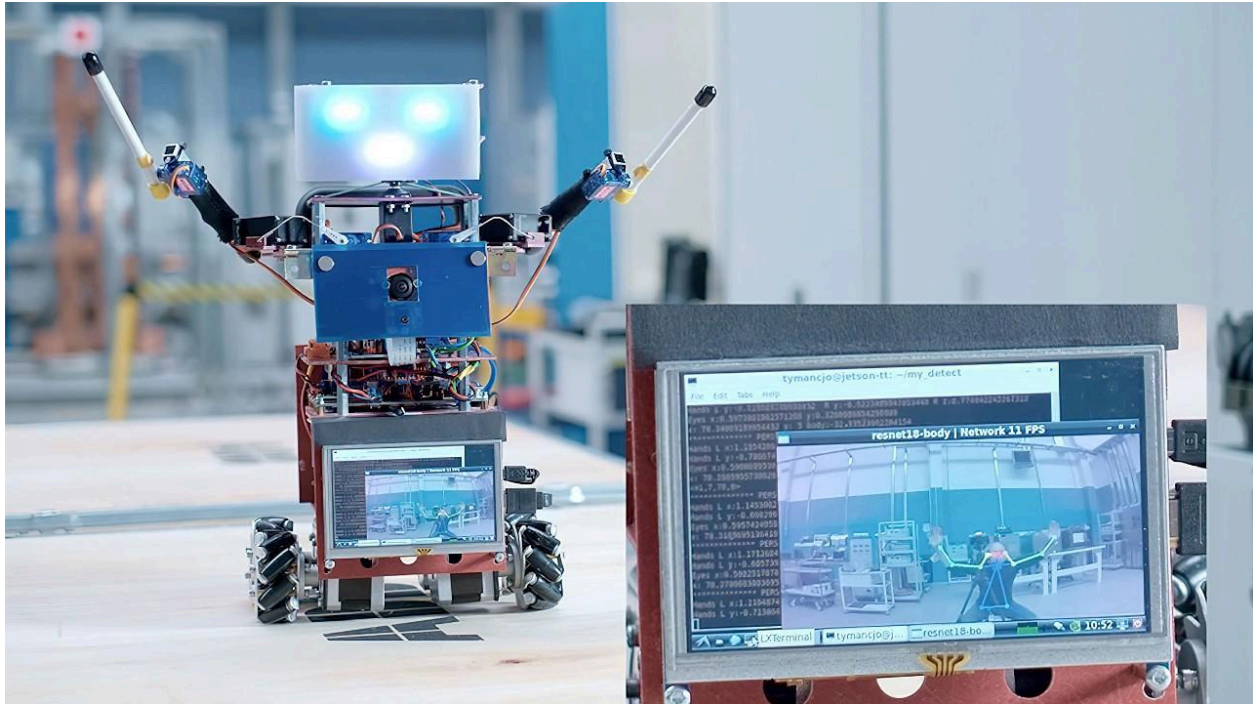
- Embedded C
 - C++
-

4. Jetson Nano









Best For

- AI at edge
- Computer vision

- Autonomous robots
- Drone AI

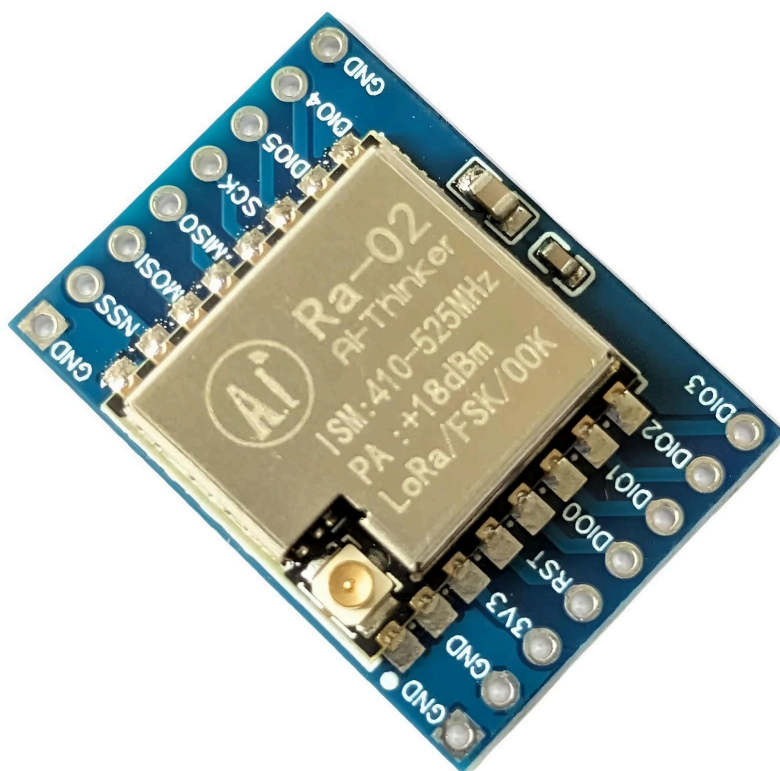
Technologies

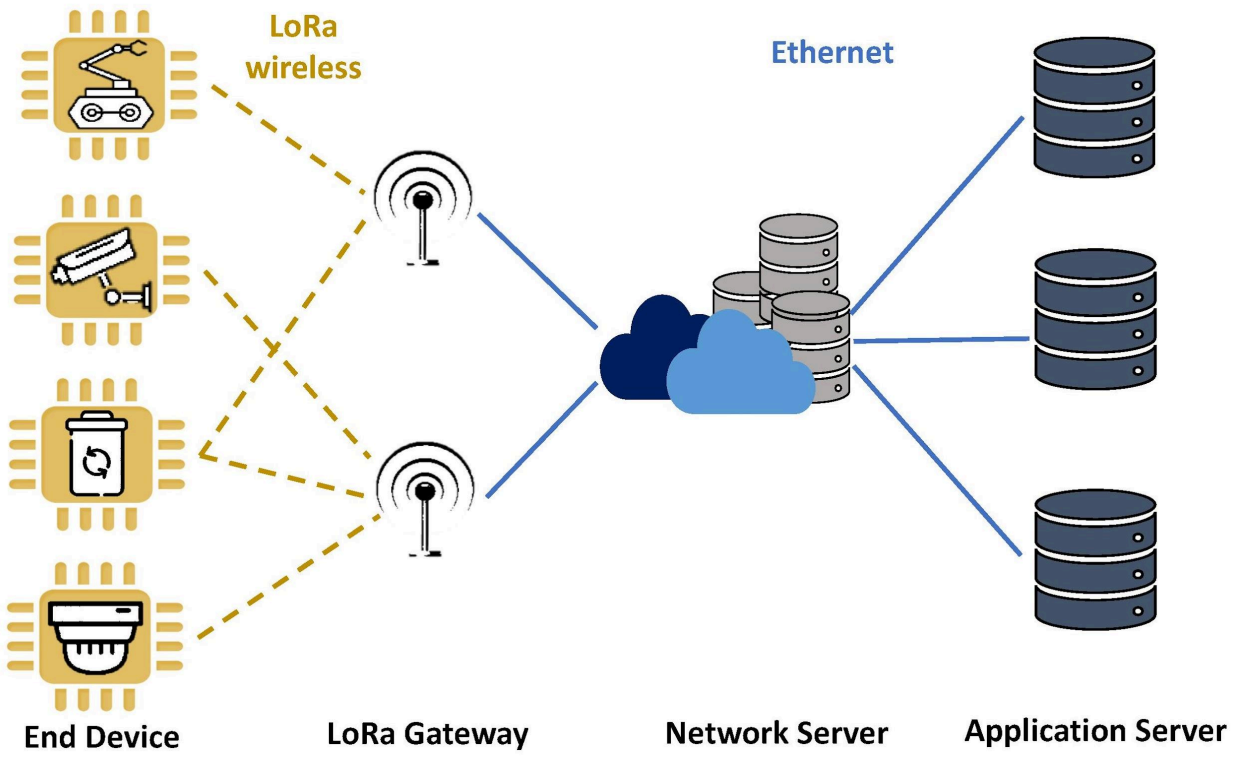
- CUDA
- TensorFlow
- PyTorch
- OpenCV

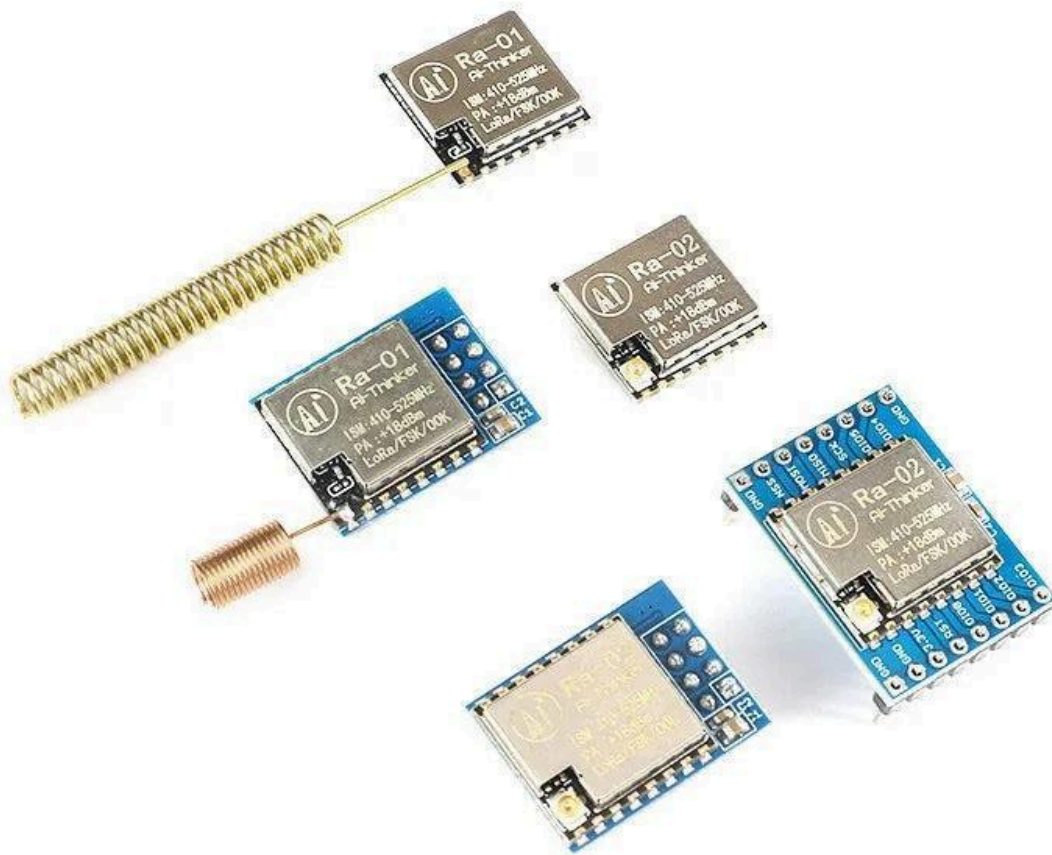
Applications

- Face recognition
 - Object detection
 - AI robotics
-

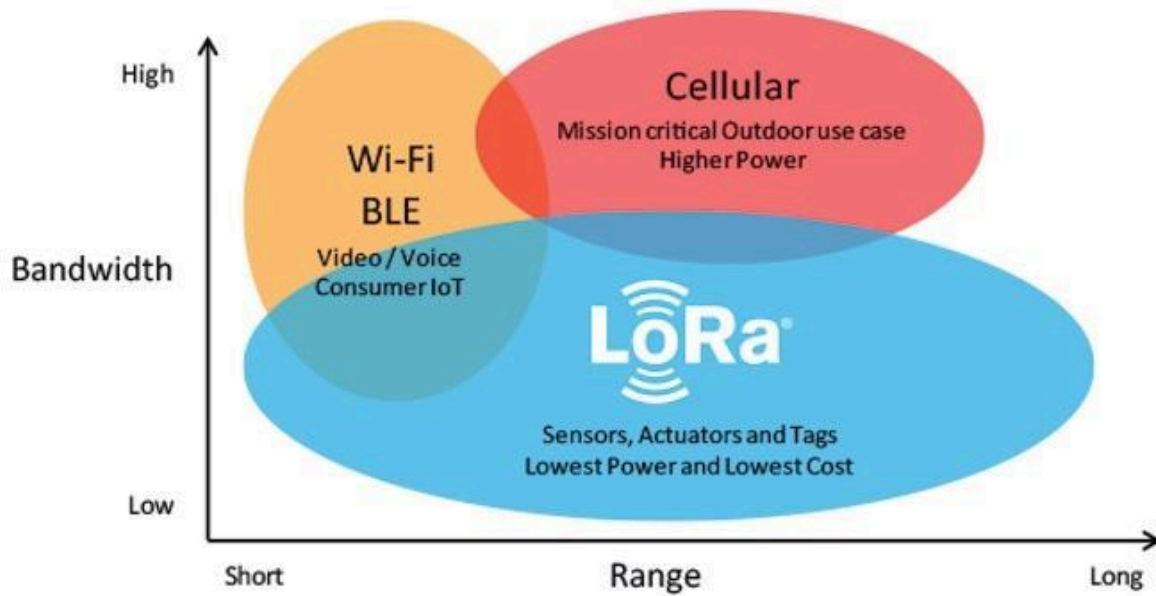
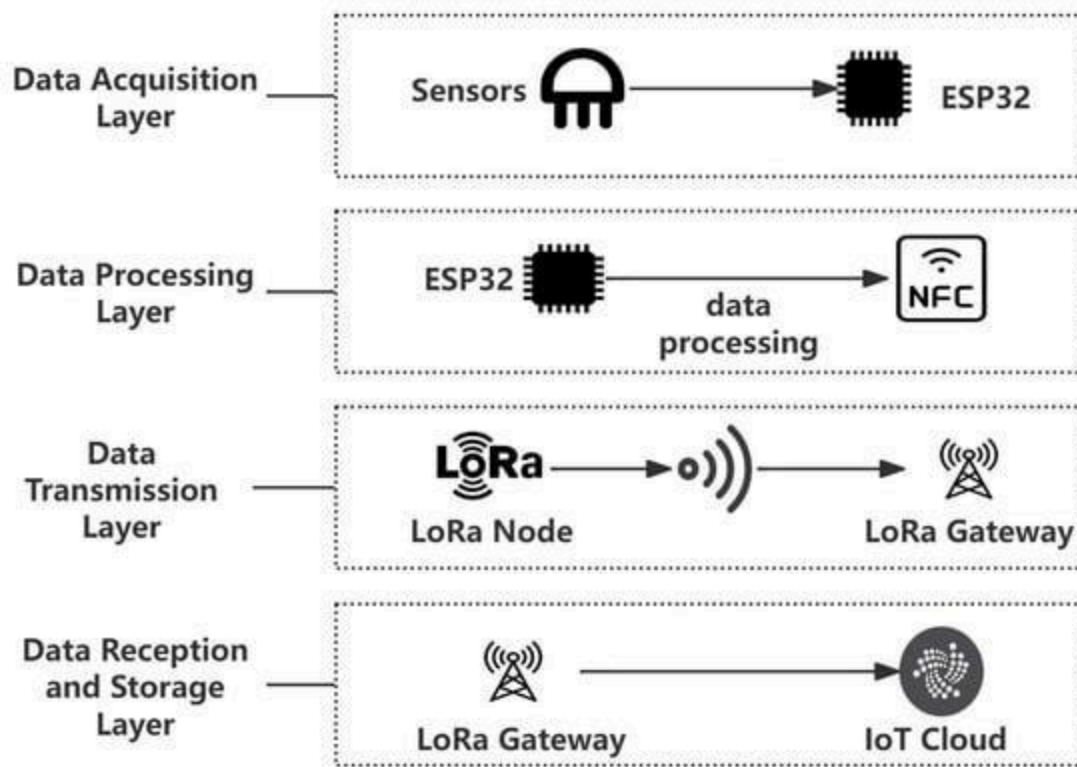
5. LoRa Modules







SX1278 Ra-02/01





Features

- Long-range communication
- Low power
- Works in remote areas

Uses

- Smart agriculture
- Defense communication
- Remote monitoring
- Sensor networking

Popular Modules

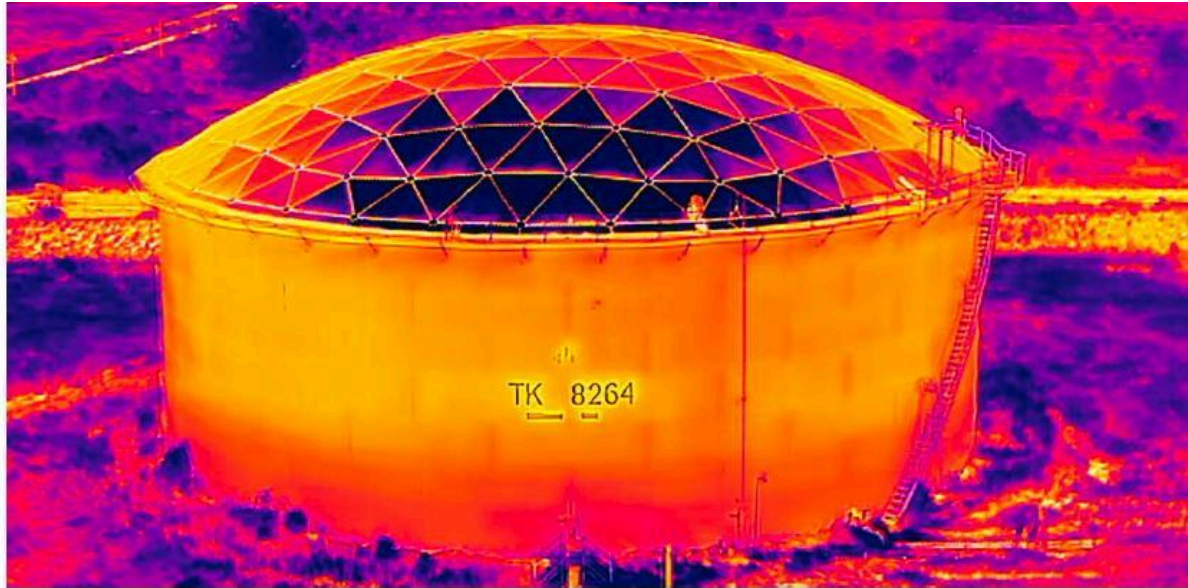
- SX1278
- Ra-02
- E32 LoRa

6. Thermal Cameras





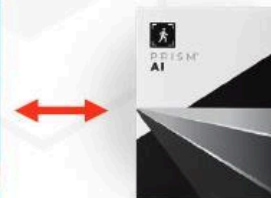


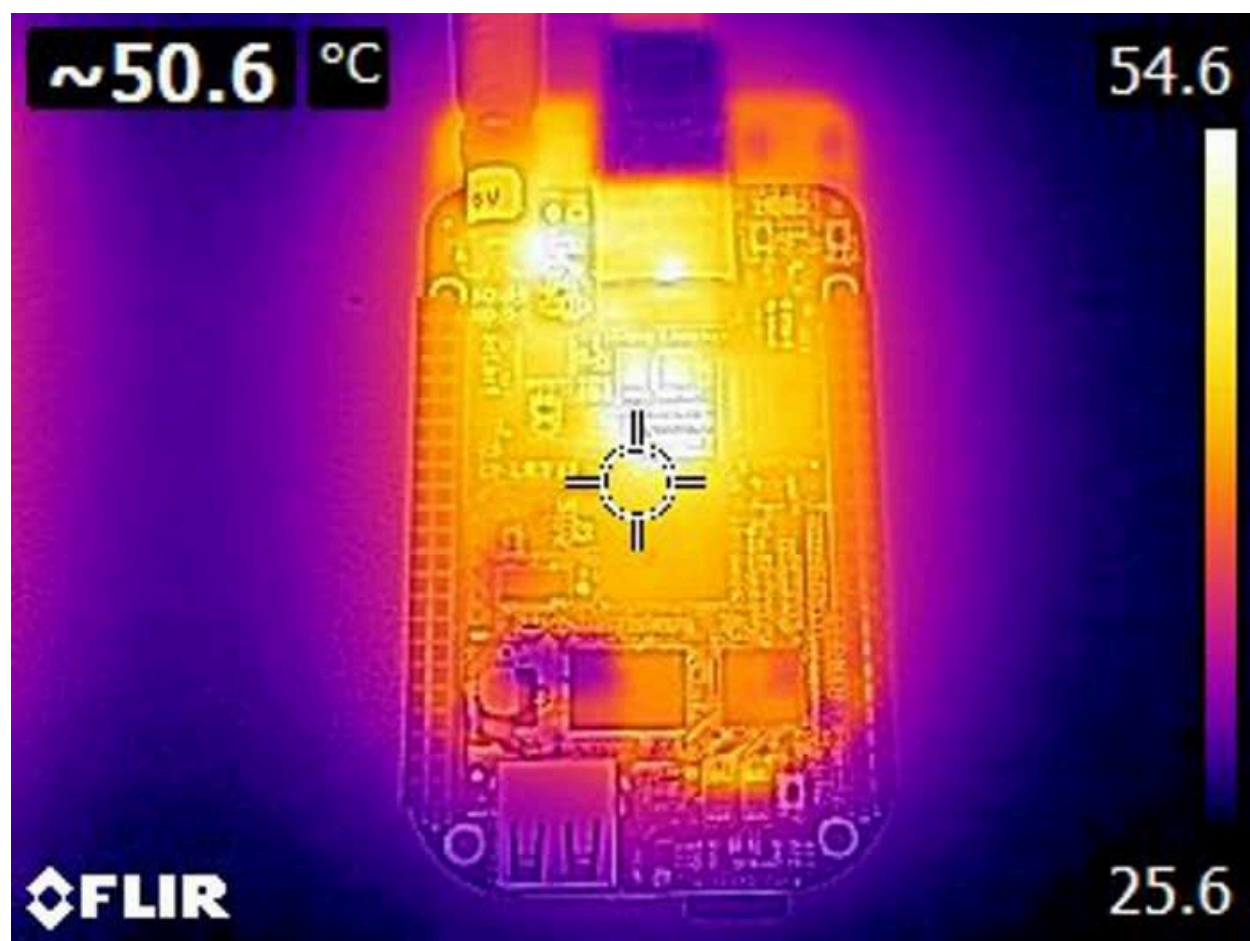


Power
MIPI-CSI
I²C



Carrier Board







Applications

- Human detection
- Night surveillance
- Industrial monitoring
- Defense systems

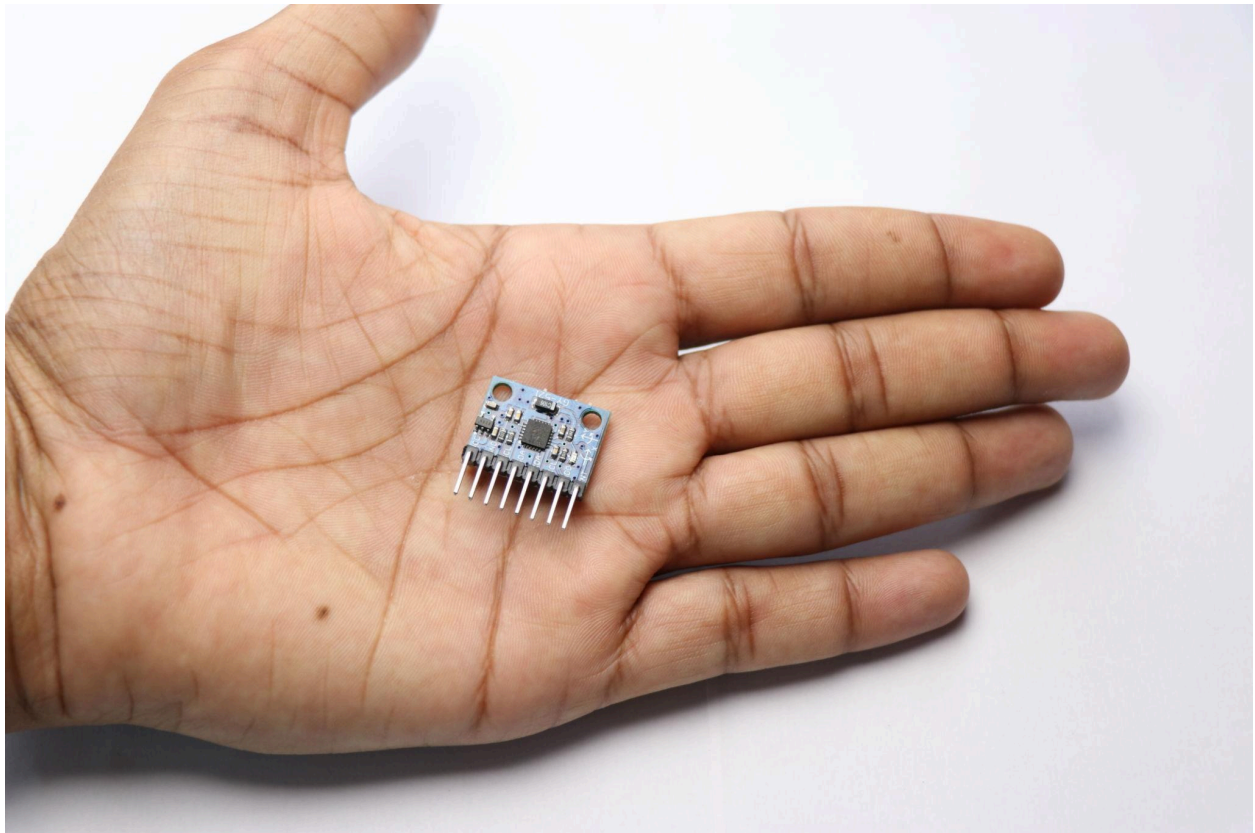
Popular Modules

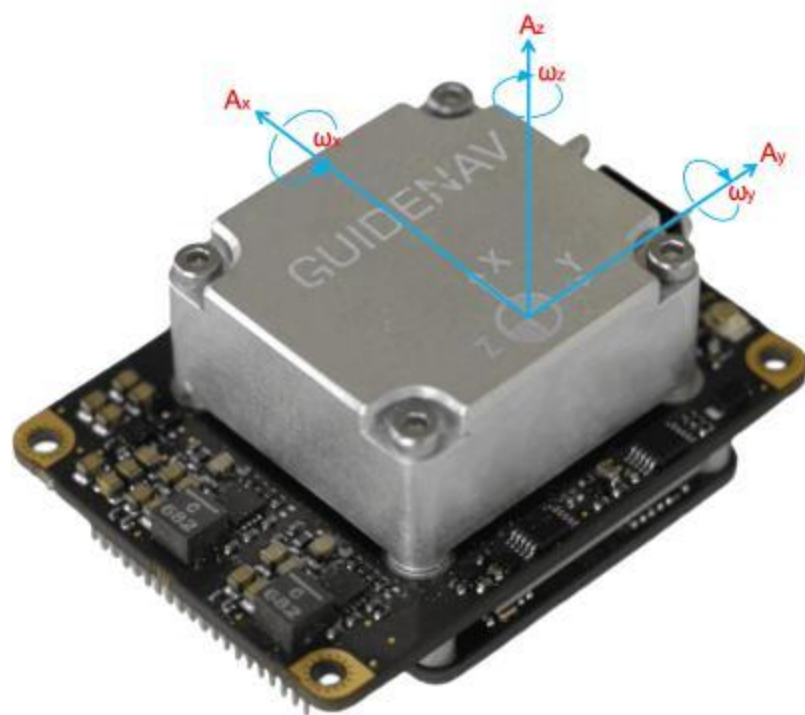
- MLX90640
- FLIR Lepton
- AMG8833

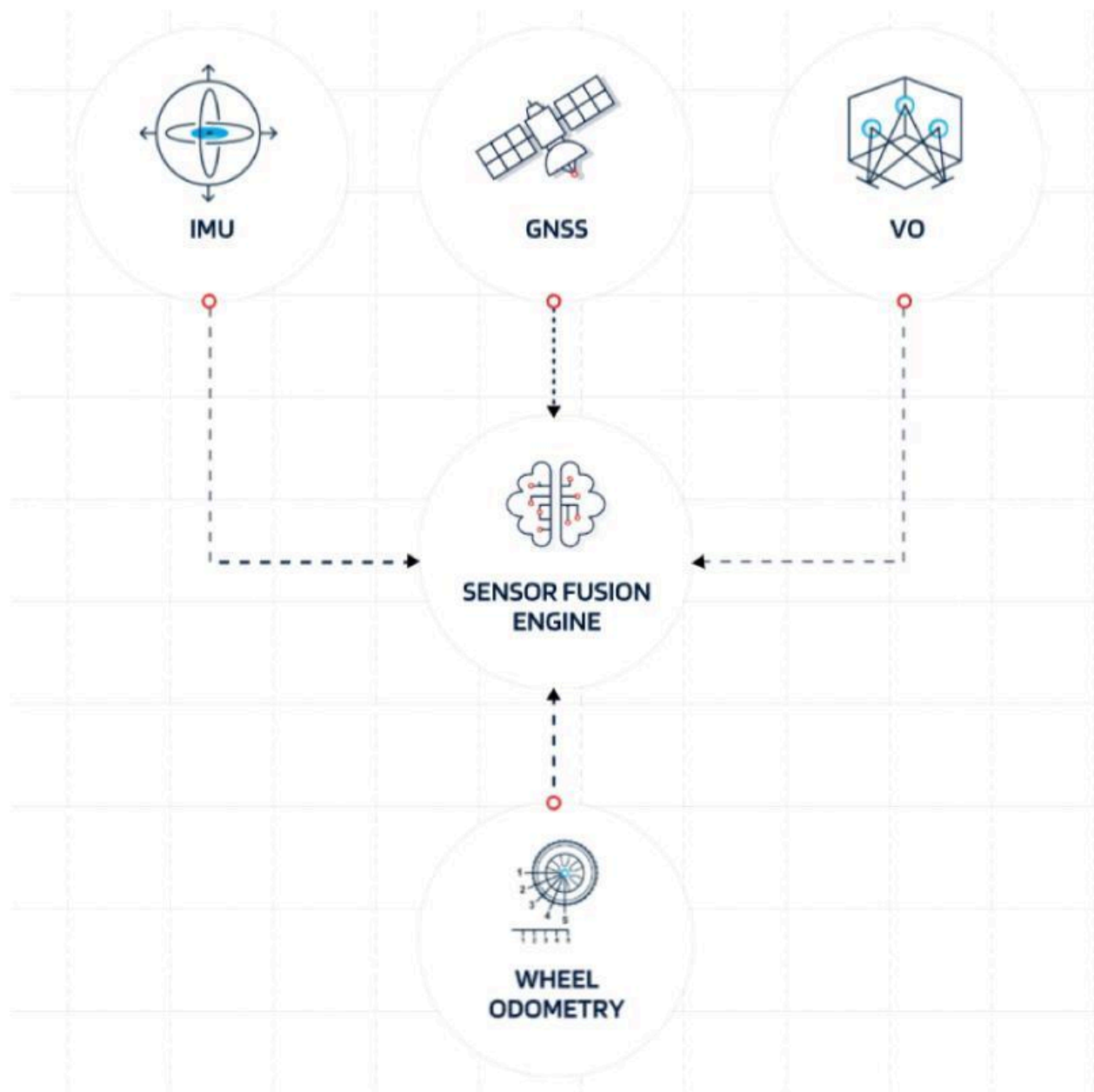
Skills

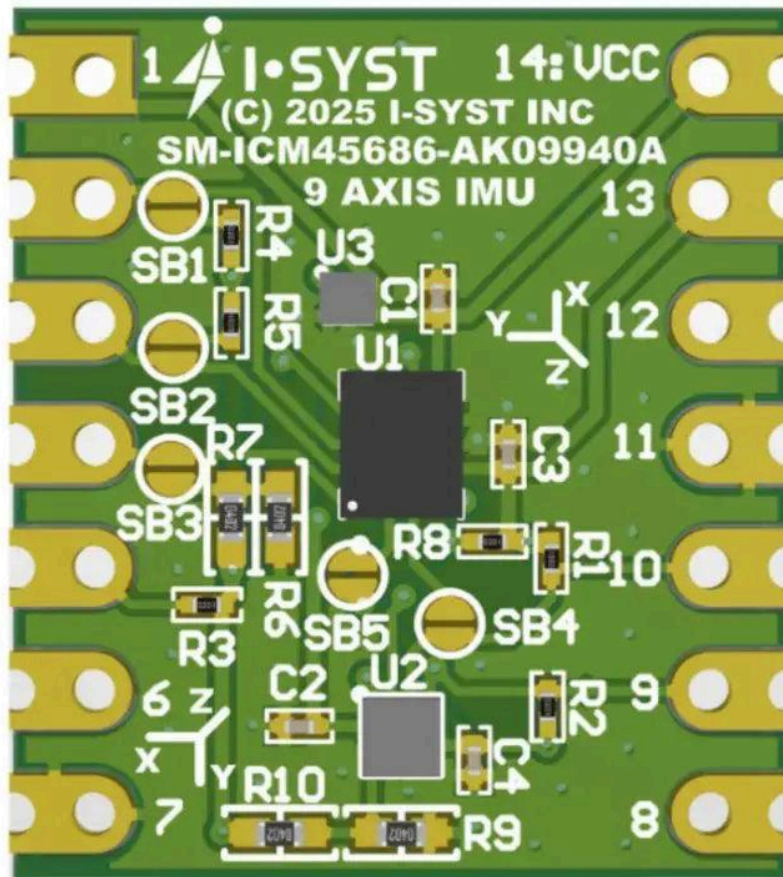
- Thermal image processing
 - AI detection
 - Heat mapping
-

7. IMU Sensors

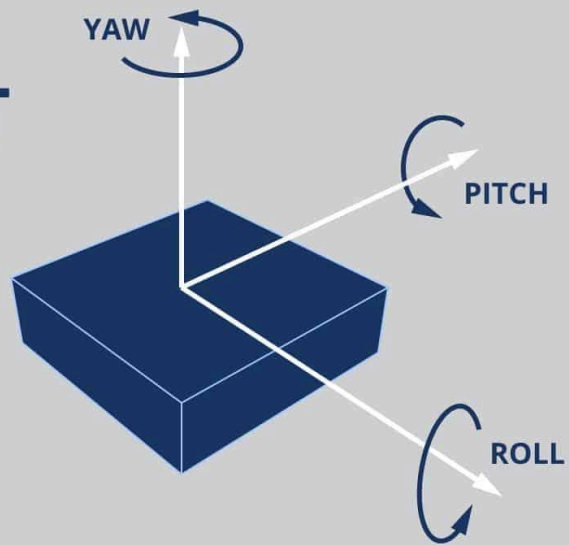








UAV INERTIAL MEASUREMENT UNITS



What IMU Measures

- Acceleration
- Rotation
- Orientation

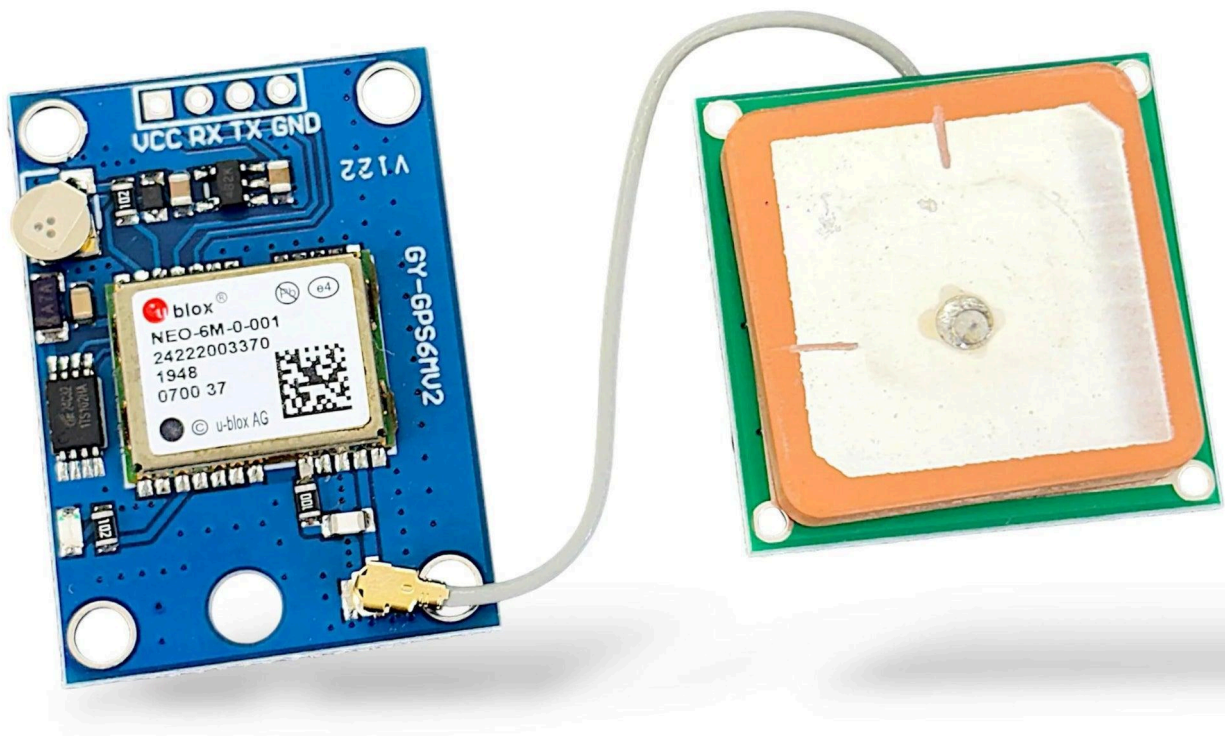
Uses

- Drone stabilization
- Robotics
- Motion tracking
- Autonomous navigation

Popular IMUs

- MPU6050
 - BNO055
 - MPU9250
-

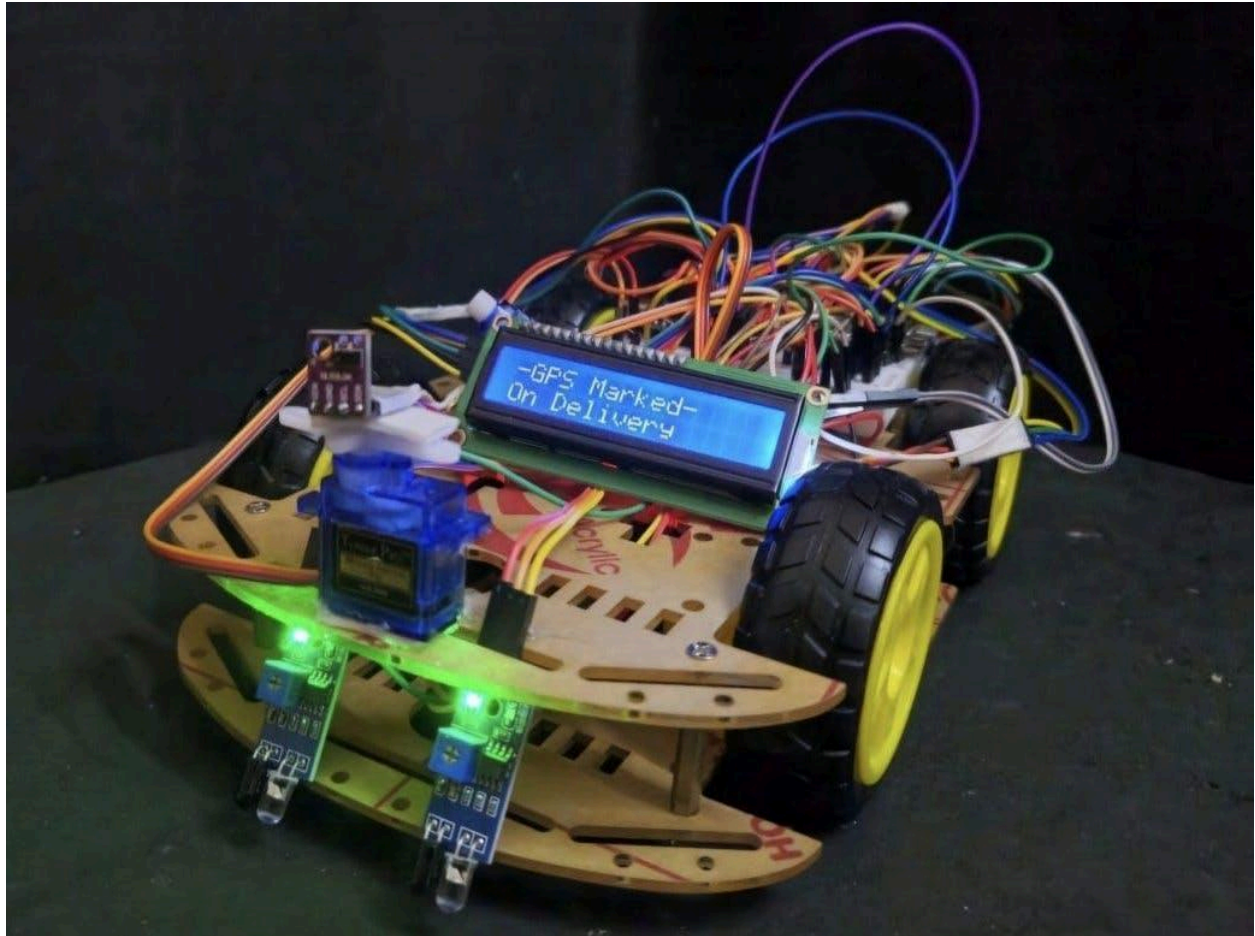
8. GPS Modules











Uses

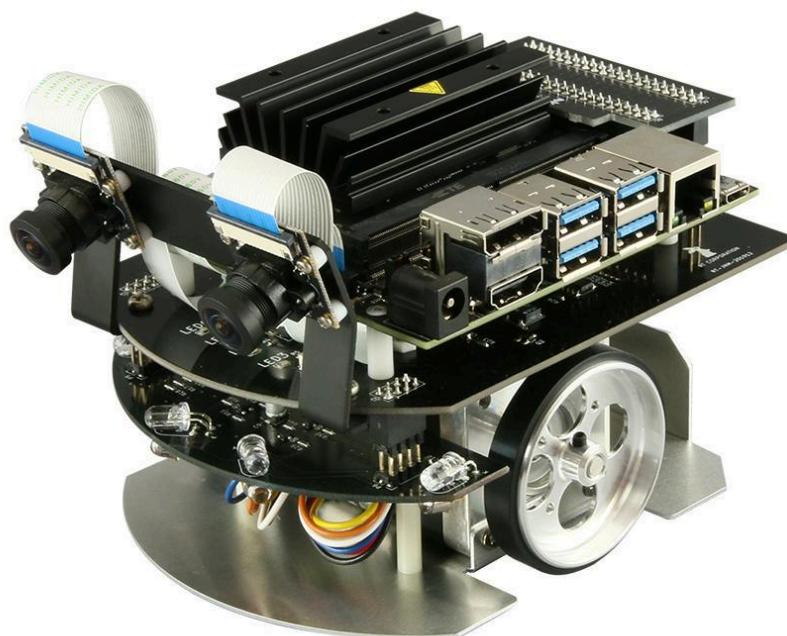
- Tracking systems
- Drone navigation
- Autonomous robots
- Vehicle monitoring

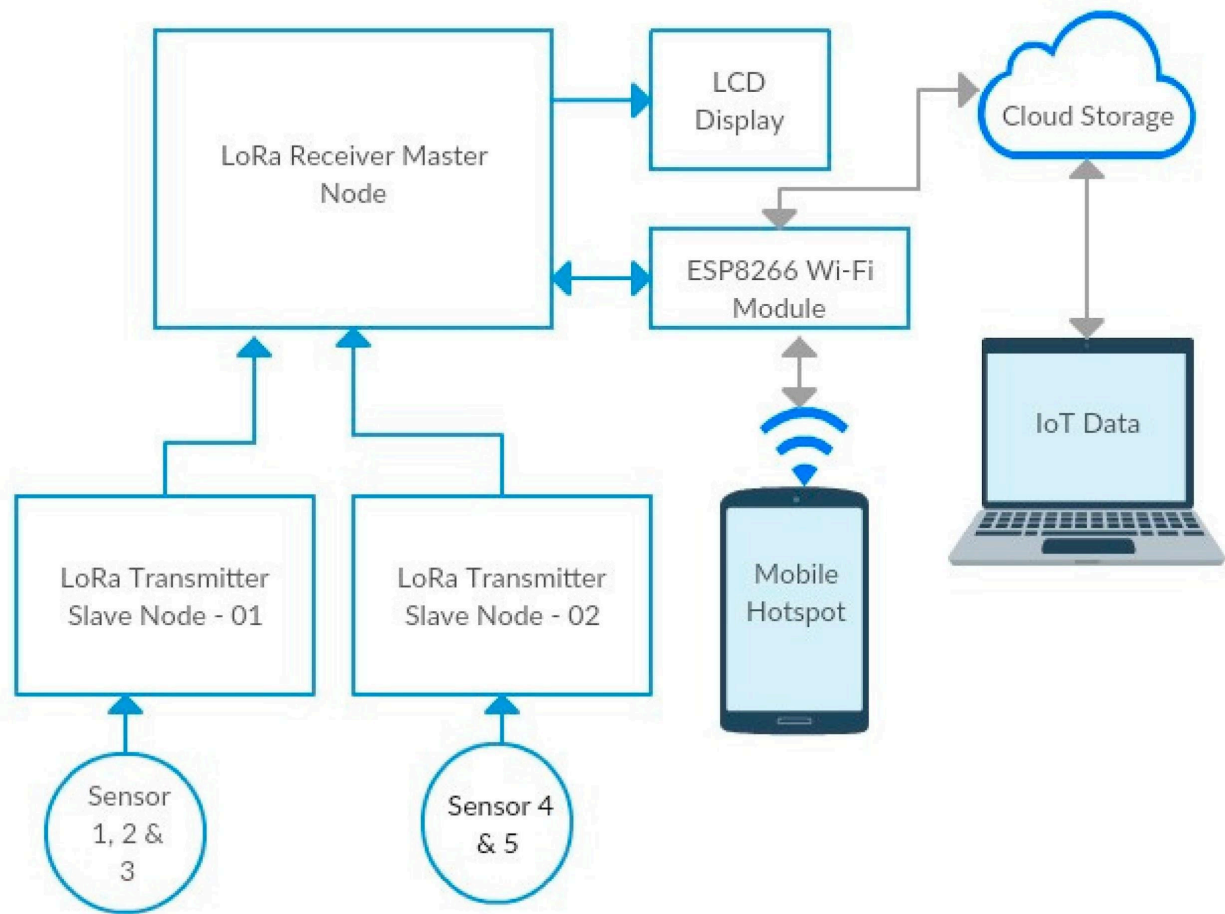
Popular Modules

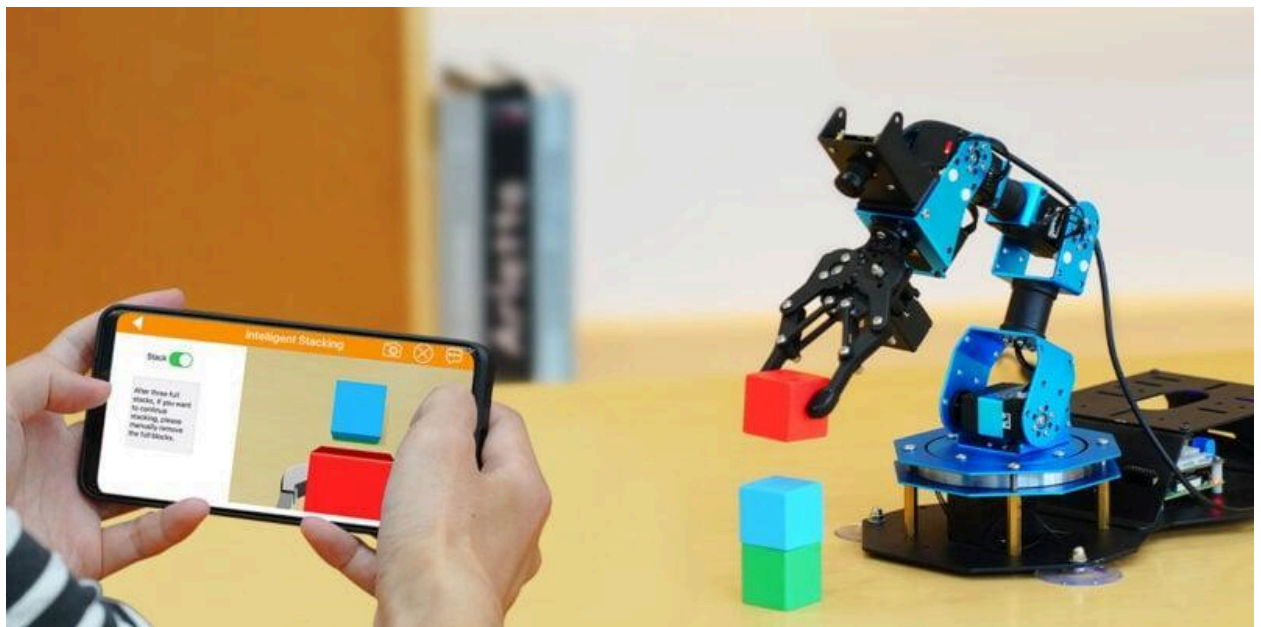
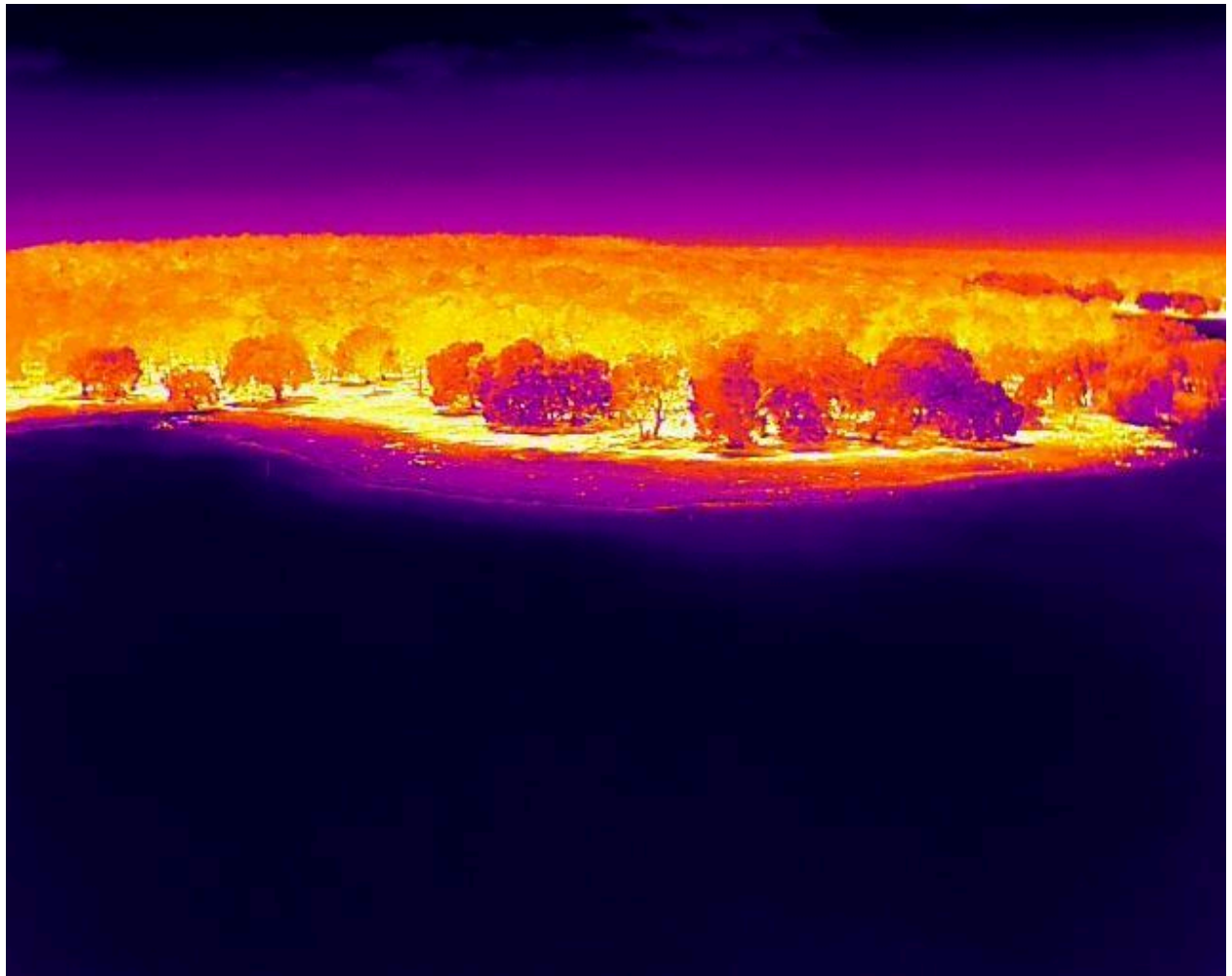
- NEO-6M
- M8N GPS
- Quectel GPS

Ultimate Embedded + AI Combo









Hardware	Main Purpose
Raspberry Pi	AI + Linux + Vision
ESP32	IoT + Wireless
STM32	Real-time control
Jetson Nano	Edge AI
LoRa	Long-range communication
Thermal Camera	Heat detection
IMU	Motion sensing
GPS	Navigation

Powerful Project Combinations

AI Surveillance Drone

- Jetson Nano
- Thermal Camera
- GPS
- IMU
- LoRa

Autonomous Rover

- Raspberry Pi
- STM32
- GPS
- Ultrasonic Sensors

Smart Defense System

- Thermal Camera
- LoRa
- AI Detection

- ESP32

Industrial AI Monitoring

- STM32
- Sensors
- Raspberry Pi
- Cloud Dashboard